Capstone Project Presentation

Wazuh Lab: Part 1 & 2 Alert Detection and Investigation & Malware Detection

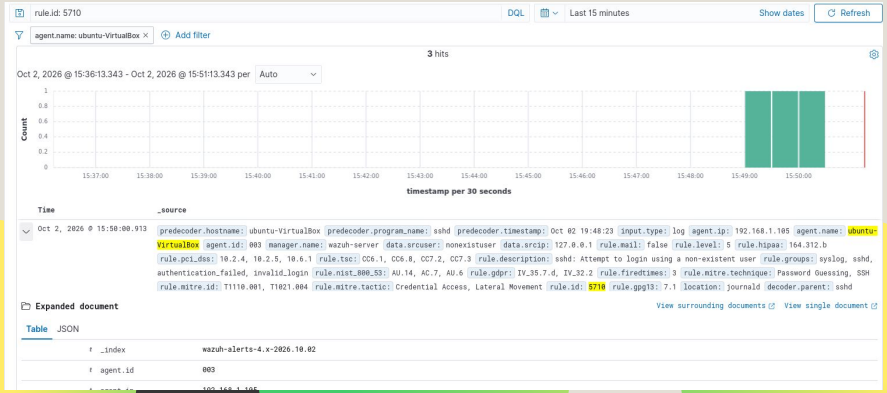
Section Focus

Part One: Alert Detection & Investigation

Detailed analysis of Wazuh security telemetry and SIEM alerts

Year/Quarter/Month

Activity 1: Failed SSH Login Attempt



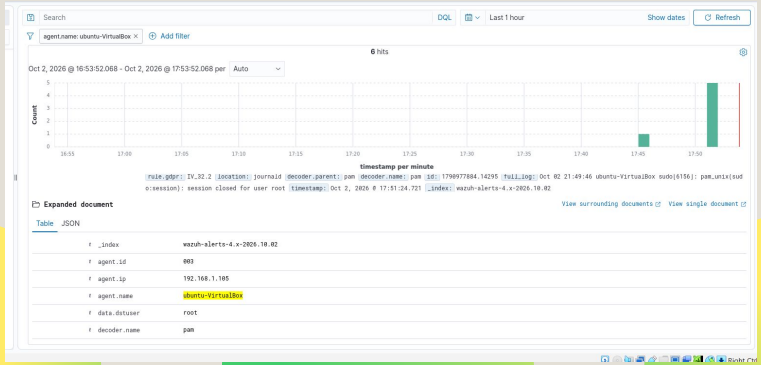
Wazuh Alert Parameters

- **Timestamp:** October 2, 2026 at 15:50:00.913
- **Rule ID:** 5710 (Severity: 5)
- **Agent:** ubuntu-virtualbox
- **Involved User:** Non-existent user
- **MITRE ATT&CK:** T1110.001 (Brute Force)

- **Analyst Explanation**

An unknown user attempted a brute force attack (T1110.001) on the Ubuntu terminal end user by inputting the wrong password credentials, and Wazuh caught the attack, which set off the 5710 (Rule ID) and level five for the severity. The attacker tried to log into the SSH 22 on Ubuntu, which failed because the account is a non-exist user, which means there is no account for the user or the account does not exist at all.

Activity 2: Failed Sudo Execution



Wazuh Alert Parameters

- **Activity:** Failed sudo execution / PAM auth failure
- **Timestamp:** October 2, 2026 at 17:51
- **Rule ID / Level:** 5502 (Level 3 - Low Severity)
- **Agent:** ubuntu-virtualbox
- **User Involved:** Non-existent user
- **MITRE ATT&CK:** T1548.002

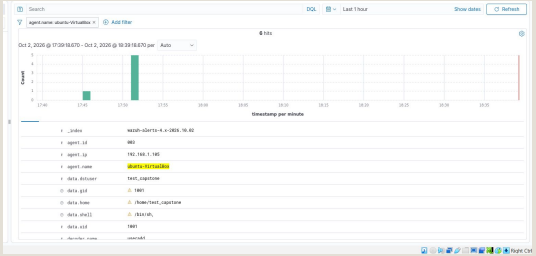
Analyst Explanation

I ran a command in ubuntu using a non-existent username in pam Linux security and security stopped the command. This is a low priority attempt because the user was trying to gain administrative privileges, which is a level 3 5502.

Activity 3: New User Account Creation

Wazuh Alert Parameters

- Activity: User account creation / persistence
- Timestamp: October 2, 2026 at 18:32:47.951
- Rule ID / Level: 5902 (Level 3 - Low Severity System Change)
- Agent: ubuntu-virtualbox
- User Created: test_capstone
- MITRE ATT&CK: T1136.001



Analyst Explanation

An attacker created a local user account called test_capstone to create administrative privileges after accessing the ubuntu terminal. The attacker used technique T1136.001 miter attack to attempt to gain access. This permission would give persistent accessing. However, Wazuh detected this administrative attempt and logged the privilege as severity level 3 and the rule ID 5902 was generated.

Part Two: Malware Detection & Response with VirusTotal

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Integrating File Integrity Monitoring (FIM) and VirusTotal API for automated threat intelligence

VirusTotal & Wazuh Integration Steps

FIM & API Parameters

- Integration: Wazuh Manager & VirusTotal API
- Monitoring: File Integrity Monitoring (FIM) on target directory
- Action: Automated file hashing and threat lookup
- Agent: ubuntu-virtualbox

Wazuh monitors and detects malicious directories and files, generating a cryptographic hash. Once that file is identified and hashed, it is sent off to VirusTotal. Virus Total then compares it against know signature files to determine if the file is malicious, assigning it a threat score. That is exactly what happened in this case. VirusTotal successfully determined that the file was malicious.

Capstone Project Complete